

Introduction to Data Engineering





Agenda

- What is Data?
- Why Data is Important?
- What is Data Engineering?
- Data Engineering Lifecycle
- Different Roles & Titles in Data Engineering
- ETL (Extract, Transform, Load) Process
- Types of Data
- Storage in Data Engineering
- Uber Use Case
- Tools & Technologies
- Course Modules



What is Data?

Data is a collection of facts, information, and statistics and this can be in various forms such as numbers, text, sound, images, or any other format.



What could be the data these companies have?

Facebook

Netflix

Spotify

Noon

Banks

Vodafone

Any seller company



NETFLIX





Company	Types of Data Collected
Facebook (Meta)	User profiles, posts, likes, friends, ads, time spent.
Netflix	Watch history, preferences, device used, search history, ratings.
Spotify	Songs played, playlists, skips, search history, ads, subscriptions.
Noon (E-commerce)	Browsing & purchase history, cart activity, payment, reviews.
Banks	Transactions, loans, card usage, fraud detection, customer support.
Seller Companies	Sales, customer details, payment trends, inventory.
Vodafone & Telecom	Calls, SMS, internet usage, location, billing, customer support.

Data is the new oil





Why Data is Important?

- 1) Decision-making
- 2) Problem solving
- 3) Understanding
- 4) Improving processes
- 5) Understanding customers



What is Data Engineering?

Data engineering is the practice of designing and building systems for the aggregation, storage and analysis of data at scale. Data engineers empower organizations to get insights in real time from large datasets.



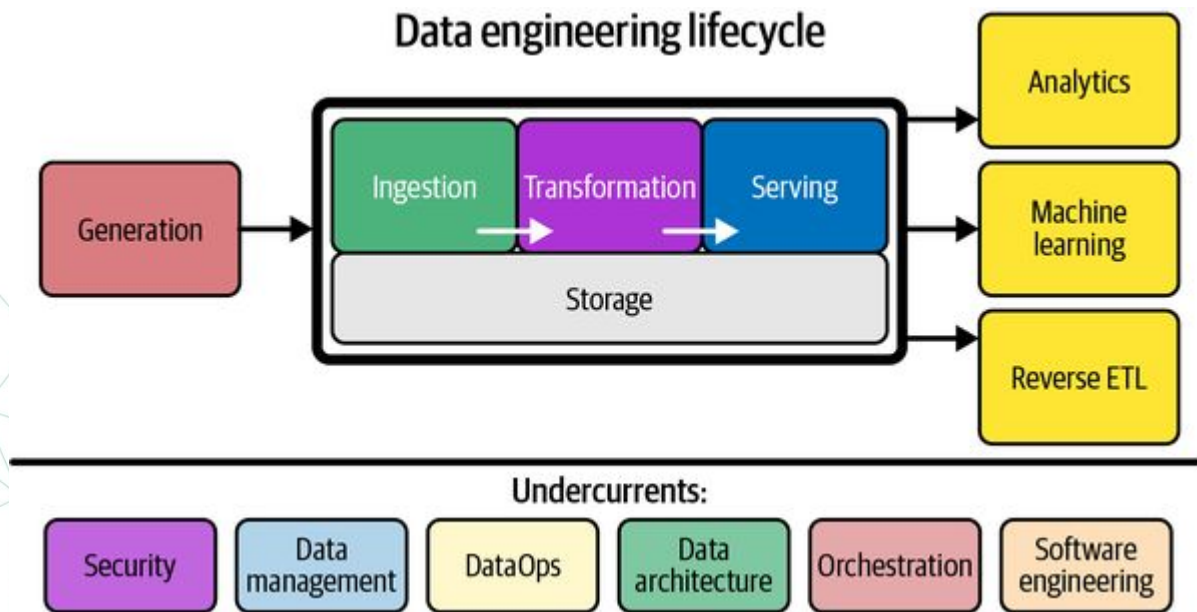


Before AI and Data Science Magic

First you need Data Engineering



Data Engineering Lifecycle

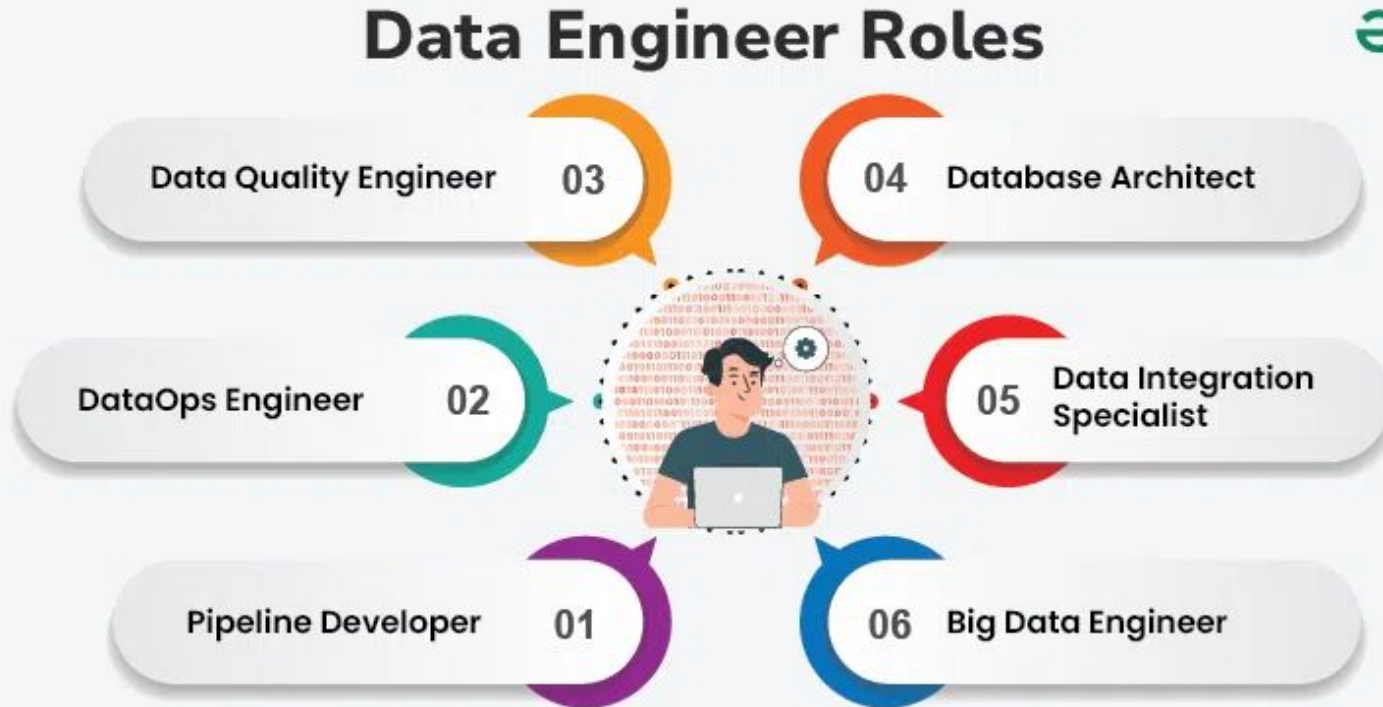




What Does a Data Engineer Do?

- **Design & Build Data Pipelines**
 - Collect, transform, and move data efficiently.
- **Develop & Manage Databases**
 - Store structured & unstructured data for easy access.
- **Ensure Data Quality & Integrity**
 - Validate data accuracy, consistency, and reliability.
- **Collaborate with Teams**
 - Work with data scientists, and analysts.
- **Optimize Data Workflows**
 - Automate processes and improve efficiency.

Different Roles



Different Titles

Data Scientist



uses statistics and machine learning to make predictions and answer key business questions

Data Engineer



build and optimize the systems that allow data scientists and analysts to perform their work

Data Analyst



deliver value by taking data, communicating the results to help make business decisions

Data Scientist

Integration

Insights

Infrastructure

Data architecture
Infrastructure setup
Database management
Scalability solutions

Modeling

Machine learning
Predictive modeling
Statistical analysis
Algorithm optimization

Reporting

KPI tracking and performance benchmarking
Reporting automation and dashboard creation
Business, market, and industry analysis

Pipeline optimization
Data cleaning automation
Real-time data processing

Data querying
Data analysis
Insight generation

Data Engineer

Jeff Winter

Data Analyst

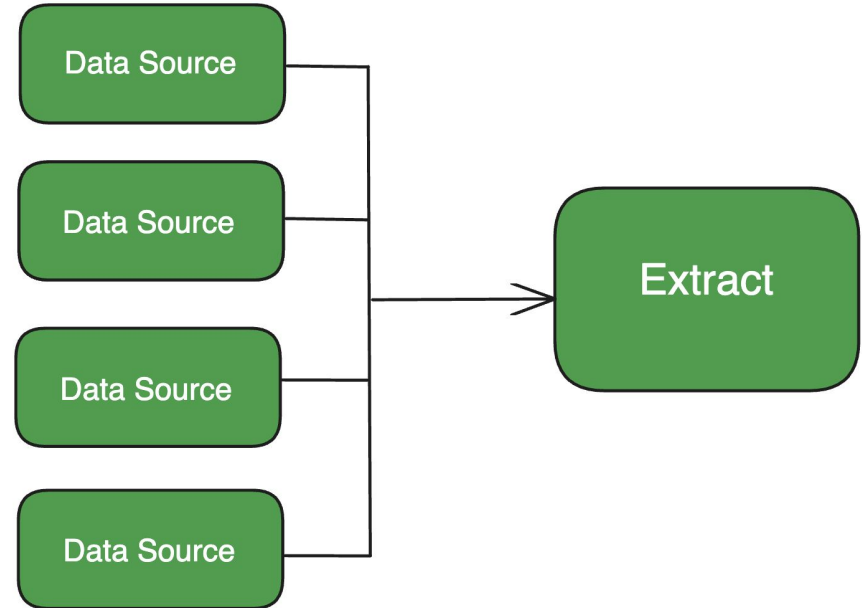
Data Engineer Closer Job: ETL

ETL stands for extract, transform, and load and is a traditionally accepted way for organizations to combine data from multiple systems into a single database, data store, data warehouse, or data lake.

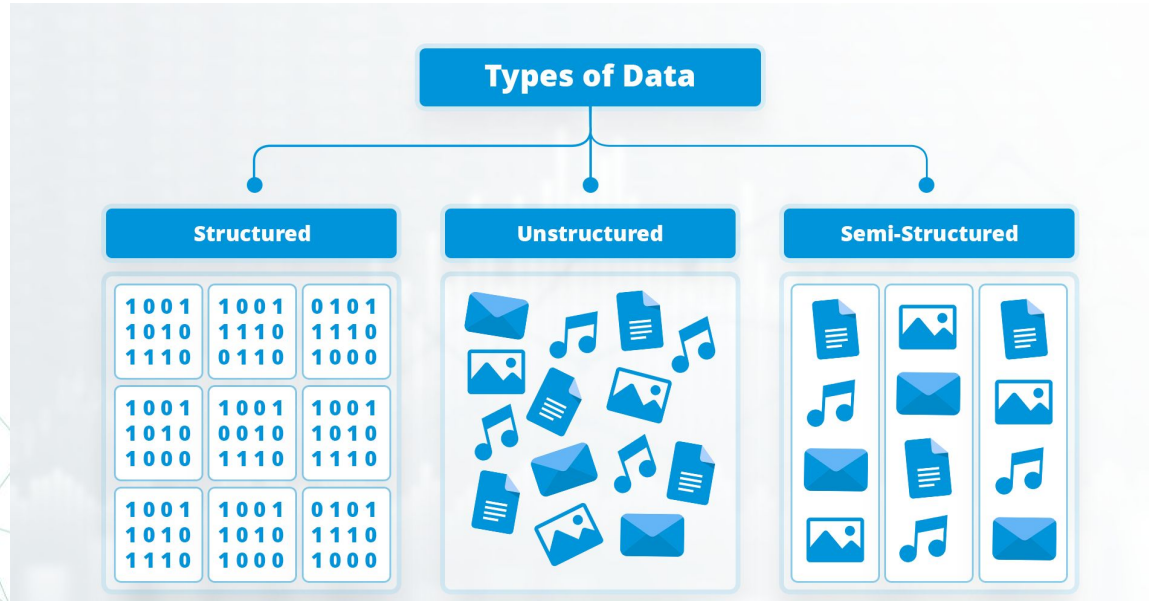


ETL: Extract

Extraction is the process of retrieving data from one or more sources—online, on-premises, legacy, SaaS, or others. After the retrieval, or extraction, is complete, the data is loaded into a staging area.



Data Sources have Several Forms





What is structured data?

Structured data is organized in a clear, predefined format. The standardized nature of structured data makes it easily decipherable by data analytics tools, machine learning algorithms and human users.

Structured data can include both quantitative data (such as prices or revenue figures) and qualitative data (such as dates, names, addresses and credit card numbers).

For example, a financial report with company names, expense values and reporting periods organized into rows and columns is considered structured data.

Example of Structured Data

index	last_name	first_name	office	address	number	city	zipcode
0	Thien	Vivian	Belgium	Martelarenlaan	38	Leuven	3010
1	Huong	Julian	Belgium	Martelarenlaan	38	Leuven	3010
2	Duplantier	Norbert	UK	Old Street	207	London	EC1V 9NR
3	McColgan	Jeff	USA	5th Ave	350	New York	10118
4	Sanchez	Rick	USA	5th Ave	350	New York	10118



What is unstructured data?

Unstructured data does not have a predefined format.

Unstructured data can contain both textual and non-textual data and both qualitative (social media comments) and quantitative (figures embedded in text) data.

Examples of unstructured data from textual data sources include, Emails, Text documents, Social media posts, Call transcripts, Message text files, such as those from Microsoft Teams or Slack

Examples of non-textual unstructured data include, Image files (JPEG, GIF and PNG), Multimedia files, Video files, Mobile activity, Sensor data from (IoT) devices



Example

Una mattina mi son alzato
O bella ciao, bella ciao, bella ciao, ciao, ciao
Una mattina mi son alzato
E ho trovato l'invasor

O partigiano, portami via
O bella ciao, bella ciao, bella ciao, ciao, ciao
O partigiano, portami via
Ché mi sento di morir

E se io muoio da partigiano
O bella ciao, bella ciao, bella ciao, ciao, ciao
E se io muoio da partigiano
Tu mi devi seppellir

E seppellire lassù in montagna
O bella ciao, bella ciao, bella ciao, ciao, ciao
E seppellire lassù in montagna
Sotto l'ombra di un bel fior

E le genti che passeranno
O bella ciao, bella ciao, bella ciao ciao ciao
E le genti che passeranno
Mi diranno «che bel fior.»

Questo è il fiore del partigiano
O bella ciao, bella ciao, bella ciao ciao ciao
Questo è il fiore del partigiano
Morto per la libertà





What is semi-structured data?

Semi-structured data has a loose structure but not a rigid schema like structured data.

Semi-structured data contains tags or markers to separate data elements but does not conform to a strict table format as in relational databases.

Examples of semi-structured data from textual data sources include:

- * JSON (JavaScript Object Notation) files
- * XML (eXtensible Markup Language) documents
- * YAML files
- * NoSQL databases (e.g., MongoDB, Couchbase)
- * Email metadata (sender, recipient, timestamp, subject)

Examples of semi-structured data from non-textual sources include:

- * Log files from web servers or applications
- * Sensor logs with labeled fields (e.g., temperature: 22.5, humidity: 70%)
- * Configuration files
- * Streaming data with nested structures (e.g., telemetry data)
- * Tagged image or video metadata (EXIF data)



Semi-structured Example

```
{
  {"user_1645156":
    "last_name": "Lacroix",
    "first_name": "Hadrien",
    "favorite_artists": ["Fools in Deed", "Gojira", "Pain", "Nanowar of Steel"]},
  {"user_5913764":
    "last_name": "Billen",
    "first_name": "Sara",
    "favorite_artists": ["Tamino", "Taylor Swift"]},
  {"user_8436791":
    "last_name": "Sulmont",
    "first_name": "Lis",
    "favorite_artists": ["Arctic Monkeys", "Rihanna", "Nina Simone"]},
  ...
}
```



ETL: Transformation

Transformation involves taking that data, cleaning it, and putting it into a common format, so it can be stored in a targeted database, data store, data warehouse, or data lake. Cleaning typically involves taking out duplicate, incomplete, or obviously erroneous records.

Example: Raw Data (Before Transformation) Vs. After

Order ID	Customer Name	Date	Amount	Country
101	John Doe	01/15/2024	\$50.00	USA
102	Jane Smith	15-01-2024	\$75.50	UK
103	NULL	2024-01-16	\$40.00	USA
104	Mark Lee	16-Jan-24	-\$20.00	CAN

Order ID	Customer Name	Date	Amount	Country
101	John Doe	2024-01-15	50.00	USA
102	Jane Smith	2024-01-15	75.50	UK
104	Mark Lee	2024-01-16	20.00	Canada



Transformation Steps Taken:

- Convert date formats to a standard format (YYYY-MM-DD)
- Remove duplicates and NULL values
- Convert negative amounts to absolute values (if applicable)
- Standardize country names



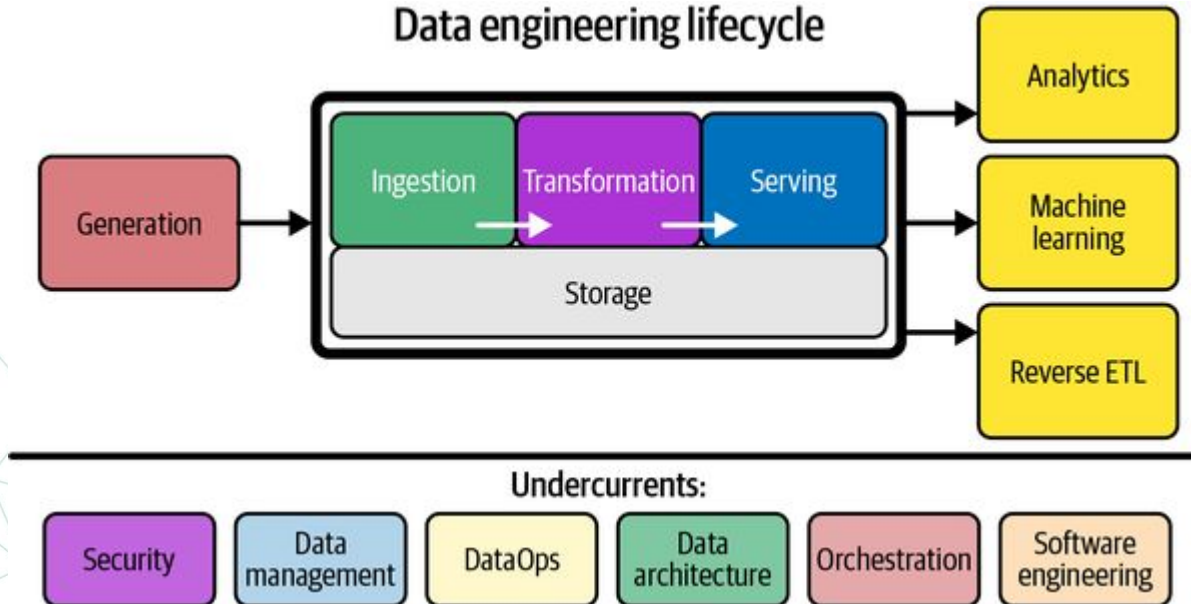
ETL: Load

Loading is the process of inserting that formatted data into the target database, data store, data warehouse, or data lake.

Scenario

A company collects daily sales data from multiple branches. After extracting the data and transforming it (cleaning, formatting), it needs to be **loaded** into a centralized “database” for reporting.

Back to the lifecycle: **Storage**

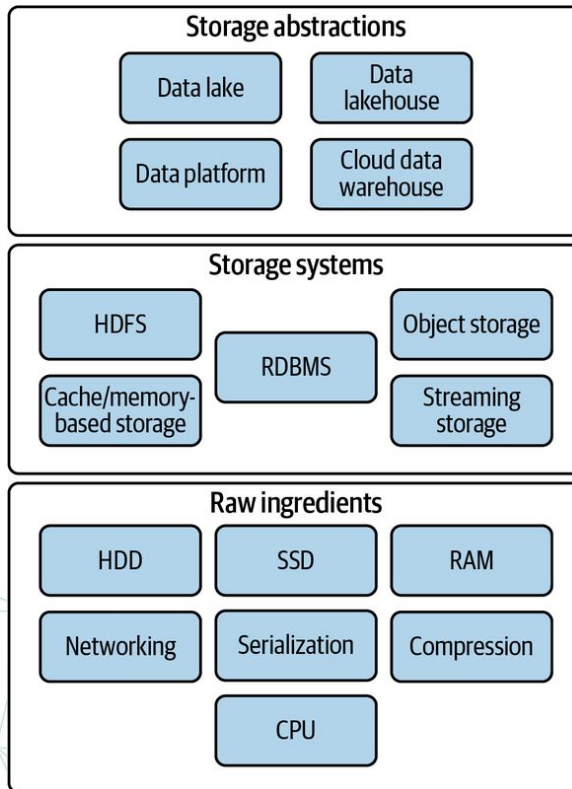




Storage

Storage is the cornerstone of the data engineering lifecycle and underlies its major stages—ingestion, transformation, and serving. Data gets stored many times as it moves through the lifecycle. To paraphrase an old saying, it's storage all the way down.

Storage Components



Different Types of Storage

Table 6-1. A heuristic cache hierarchy displaying storage types with approximate pricing and performance characteristics

Storage type	Data fetch latency ^a	Bandwidth	Price
CPU cache	1 nanosecond	1 TB/s	N/A
RAM	0.1 microseconds	100 GB/s	\$10/GB
SSD	0.1 milliseconds	4 GB/s	\$0.20/GB
HDD	4 milliseconds	300 MB/s	\$0.03/GB
Object storage	100 milliseconds	10 GB/s	\$0.02/GB per month
Archival storage	12 hours	Same as object storage once data is available	\$0.004/GB per month

^a A microsecond is 1,000 nanoseconds, and a millisecond is 1,000 microseconds.



Data Engineering Storage Abstractions

- Data warehouses
- Data lakes
- Data lakehouses

Comparison

Feature	Data Warehouse	Data Lake	Data Lakehouse
Data Types	Structured data only	Structured, semi-structured, unstructured	Structured, semi-structured, unstructured
Storage Cost	High (optimized for structured data)	Low (scalable for large volumes)	Moderate (balances cost & performance)
Performance	Fast for structured queries	Slower without optimization	Optimized for both structured & unstructured data

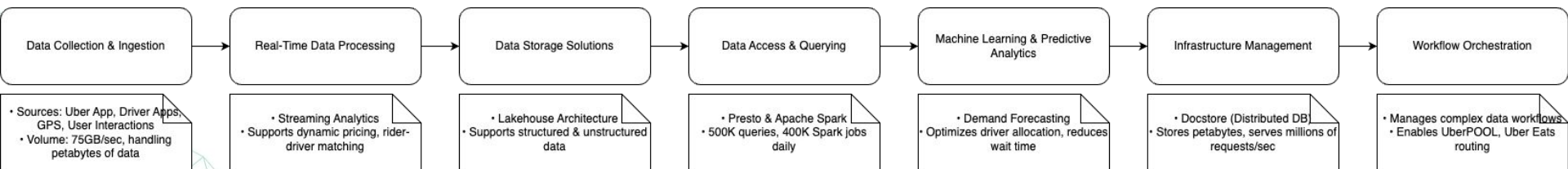
Uber use-case



Uber



Uber Abstracted Data Pipeline Flow



Assignment



Technology & Tools Used in the Course

Programming: Python

Cloud: Microsoft Azure

Storage Tools: MySQL/PostgreSQL

Querying: SQL

Processing: Apache Spark, Hadoop.

Orchestration: Apache Airflow

Course Modules Overview

Thank you